Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application - Hospital Management System(HMS)

Aim

To analyze the Software Development Life Cycle (SDLC) models used in the development of a Hospital Management System and identify the most suitable SDLC model.

Objective

- Study different SDLC models.
- Analyze the development process of a Hospital Management System.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for the application.

Selected Software Application

Hospital Management System (HMS)

Overview

A Hospital Management System is a software application that helps hospitals to manage daily operations efficiently. It allows patients,doctors,nurses,pharmacists,laboratory technicians, and administrators to access and manage information digitally.

The system provides services such as:

- Patient Registration
- Doctor Appointment Booking
- Electronic Medical Records
- Laboratory Report Management
- Pharmacy Management
- Billing and Payment
- Staff Management

Problem Statement

Many hospitals still rely on manual or semi-computerized methods to manage patient records, appointments, billing, laboratory reports, and pharmacy services.

This leads to several challenges, such as:

- Long waiting times for patients due to manual appointment scheduling.
- Difficulty in maintaining accurate patient medical records.
- Errors in billing and pharmacy management.
- Delays in accessing laboratory reports.
- Poor communication between hospital departments.
- Risk of data loss and unauthorized access to patient information.
- Time-consuming report generation and administrative tasks.

As the number of patients increases, managing hospital operations manually becomes inefficient, resulting in reduced productivity, increased operational costs, and lower quality of patient care.

Proposed Solution

A Hospital Management System (HMS) is developed to automate and integrate all major hospital operations into a single digital platform.

The proposed solution provides:

- Online patient registration and appointment booking.
- Electronic Medical Records (EMR) for secure patient information management.
- Doctor module for diagnosis, prescriptions, and appointment management.
- Laboratory management for test requests and report generation.
- Pharmacy management with medicine inventory control.
- Automated billing and payment processing.
- Secure role-based access for doctors, nurses, pharmacists, laboratory staff, and administrators.
- Real-time report generation and hospital analytics.
- Cloud-based data storage with backup and security mechanisms.

The system follows the **Agile SDLC model**, enabling continuous development, frequent testing, and regular updates based on feedback from hospital staff and administrators. This approach improves operational efficiency, reduces manual errors, enhances patient care, and ensures secure management of hospital information.

Conclusion

This case study demonstrates that selecting the appropriate SDLC model depends on the project requirements. For a Hospital Management System, Agile provides flexibility, rapid feature delivery, and continuous collaboration with doctors, hospital staff, and administrators. DevOps enhances deployment automation, system reliability, and maintenance. Together, Agile and DevOps ensure high software quality, improved patient care, efficient hospital operations, and better user satisfaction.